

Web Programming

Web Programming

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CSS first steps

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CSS — Styling the web

▼ CSS first steps

CSS first steps overview

What is CSS?

Getting started with CSS

How CSS is structured

How CSS works

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JavaScript — Dynamic client-side scripting

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▶ Introducing JavaScript objects

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Learn to style HTML using CSS

Cascading Style Sheets — or [CSS](#) — is the first technology you should start learning after [HTML](#). While HTML is used to define the structure and semantics of your content, CSS is used to style it and lay it out. For example, you can use CSS to alter the font, color, size, and spacing of your content, split it into multiple columns, or add animations and other decorative features.

Looking to become a front-end web developer?

We have put together a course that includes all the essential information you need to work towards your goal.

[Get started](#)

Prerequisites

You should learn the basics of HTML before attempting any CSS. We recommend that you work through our [Introduction to HTML](#) module first.

Once you understand the fundamentals of HTML, CSS at the same time, moving back and forth between the two is more interesting and much more fun to learn when you apply CSS, and you can't learn CSS without

In this article

Prerequisites

Modules

Solving common CSS problems

"CSS is weird"

See also



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<https://tinyurl.com/znuz8u2u>

Getting started with CSS

Lecture: Getting started with CSS

Module: CSS first steps

Introduction

Objective of the lecture

Objective: To understand the basics of linking a CSS document to an HTML file and be able to do simple text formatting with CSS.

Introduction

- In this **Lecture** we will take a **simple HTML document** and **apply CSS** to it, **learning** some **practical things** about the **language** along the way.

Starting with some HTML

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4     <meta charset="utf-8">
5     <title>Getting started with CSS</title>
6 </head>
7
8 <body>
9     <h1>I am a level one heading</h1>
10
11     <p>This is a paragraph of text. In the text is a <span>span element</span>
12     and also a <a href="http://example.com">link</a>.</p>
13
14     <p>This is the second paragraph. It contains an <em>emphasized</em> element.</p>
15
16     <ul>
17         <li>Item one</li>
18         <li>Item two</li>
19         <li>Item <em>three</em></li>
20     </ul>
21
22 </body>
23 </html>
```

Copy this code if you want to work on your own computer. Save it as `index.html` in a folder on your machine.

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

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Captain Anonymous

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HTML

```
1 <body>
2   <h1>I am a level one
   heading</h1>
3
4   <p>This is a paragraph of
   text. In the text is a <span>span
   element</span>
5   and also a <a
   href="http://example.com">link</a>.</p>
6
7   <p>This is the second
   paragraph. It contains an
   <em>emphasized</em> element.</p>
8
9   <ul>
```

CSS

1

JS

1

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

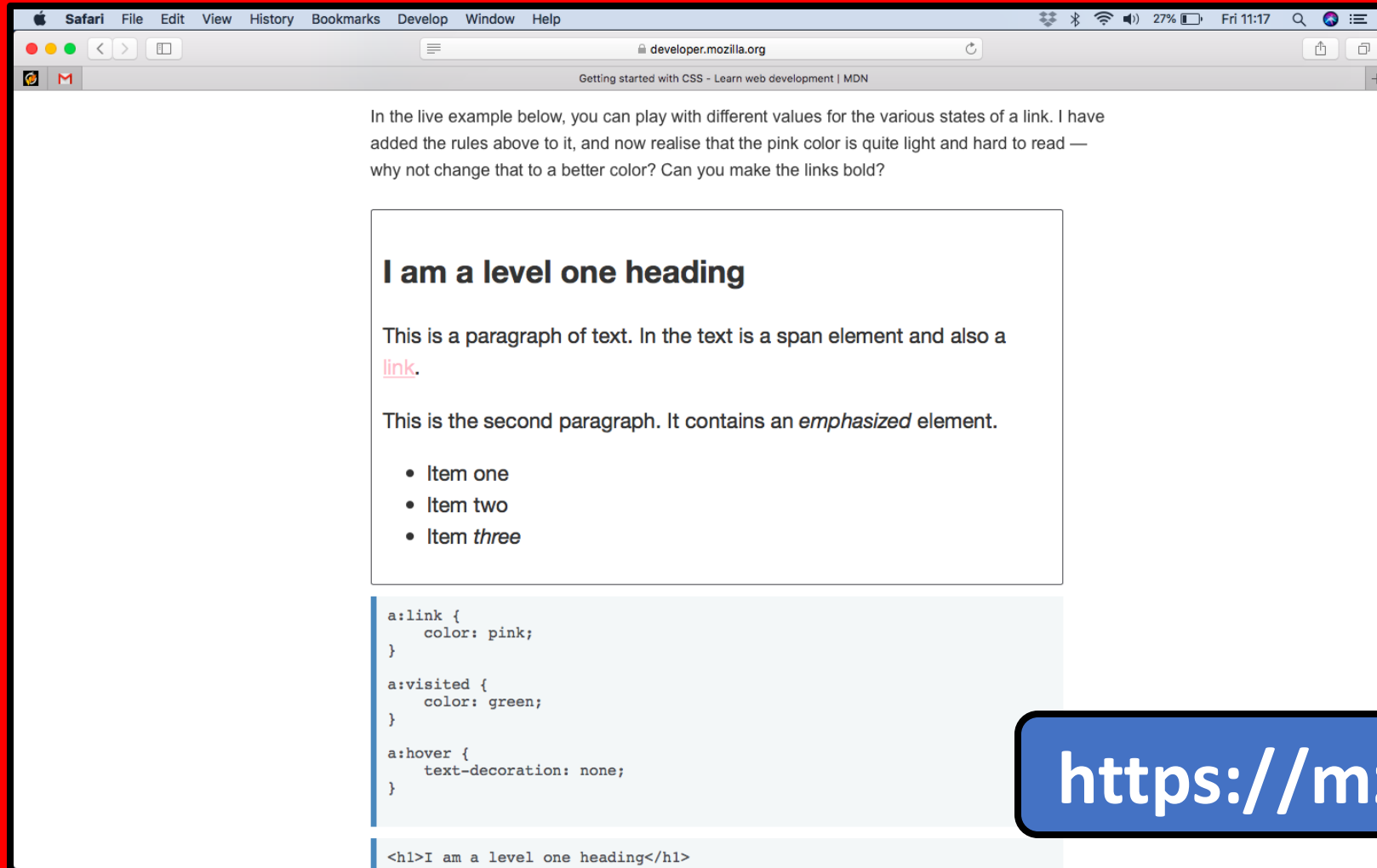
This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

<https://codepen.io>

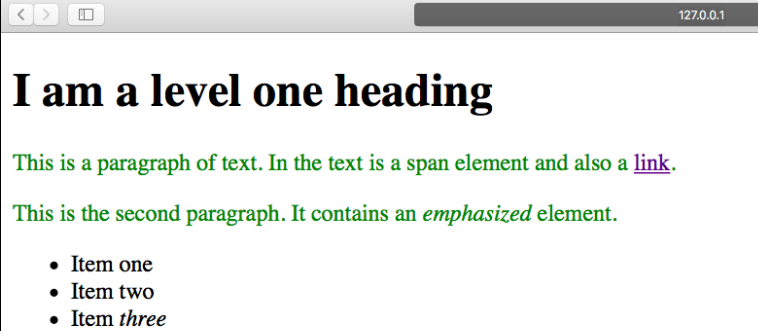
Active learning: Styling HTML elements

Go to the link indicated below for the Active Learning statement



<https://mzl.la/2EfiOtr>

Solution



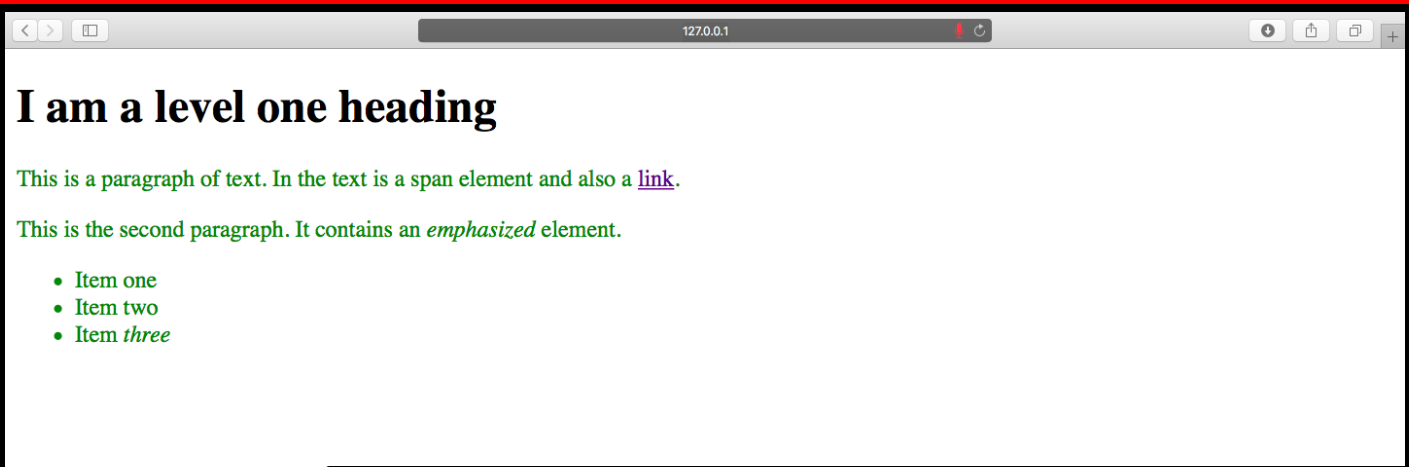
A screenshot of a web browser window. The address bar shows '127.0.0.1'. The page content includes a level one heading, two paragraphs, and a list of three items.

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*



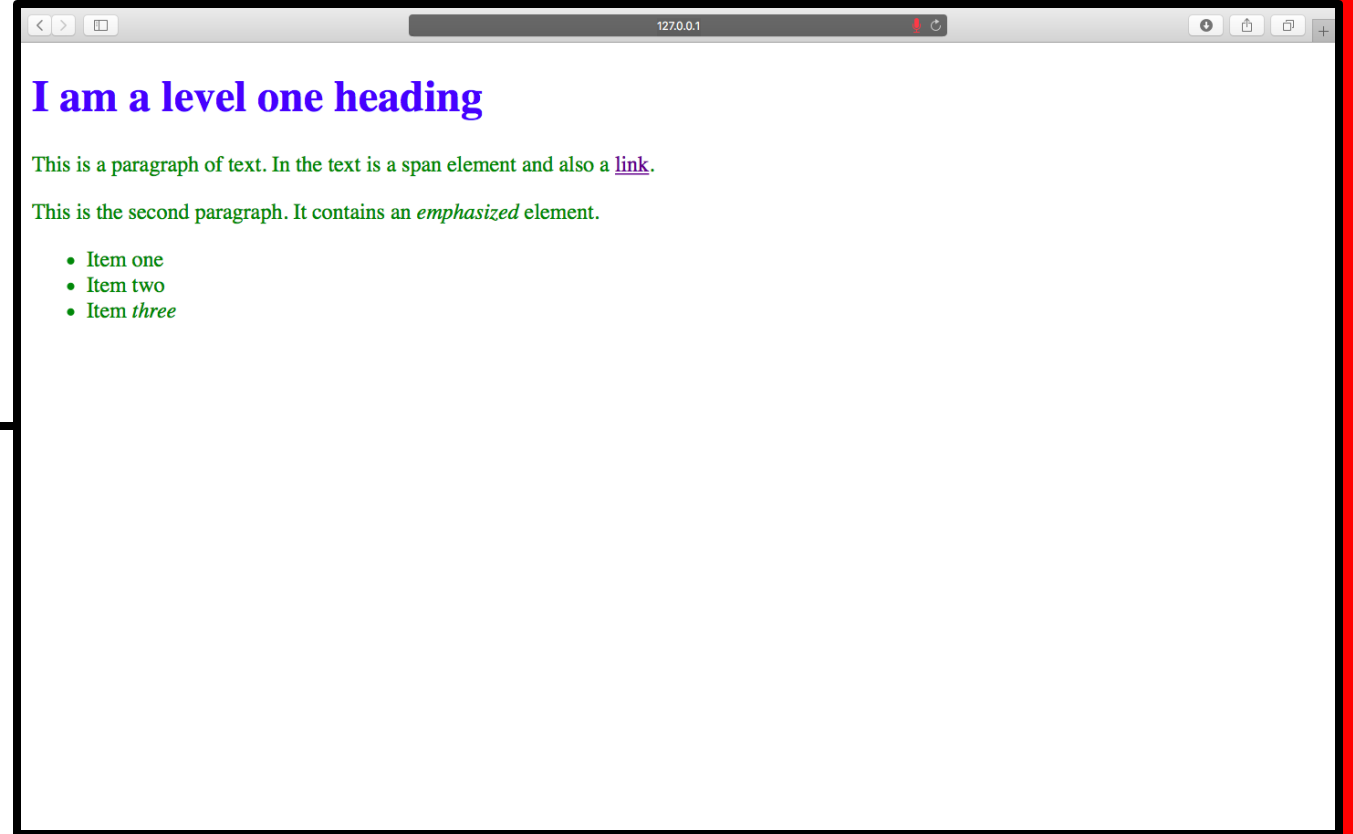
A screenshot of a web browser window. The address bar shows '127.0.0.1'. The page content includes a level one heading, two paragraphs, and a list of three items.

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*



A screenshot of a web browser window. The address bar shows '127.0.0.1'. The page content includes a level one heading, two paragraphs, and a list of three items.

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

Adding CSS to our document

Adding CSS to our document

- There are **3** different **ways** to **apply CSS** to an **HTML document** that you'll commonly **come across**.
- For now, **we will look** at the **most usual** and **useful way** of doing so.
 - ✓ **Linking CSS** from the **head** of your **document**.

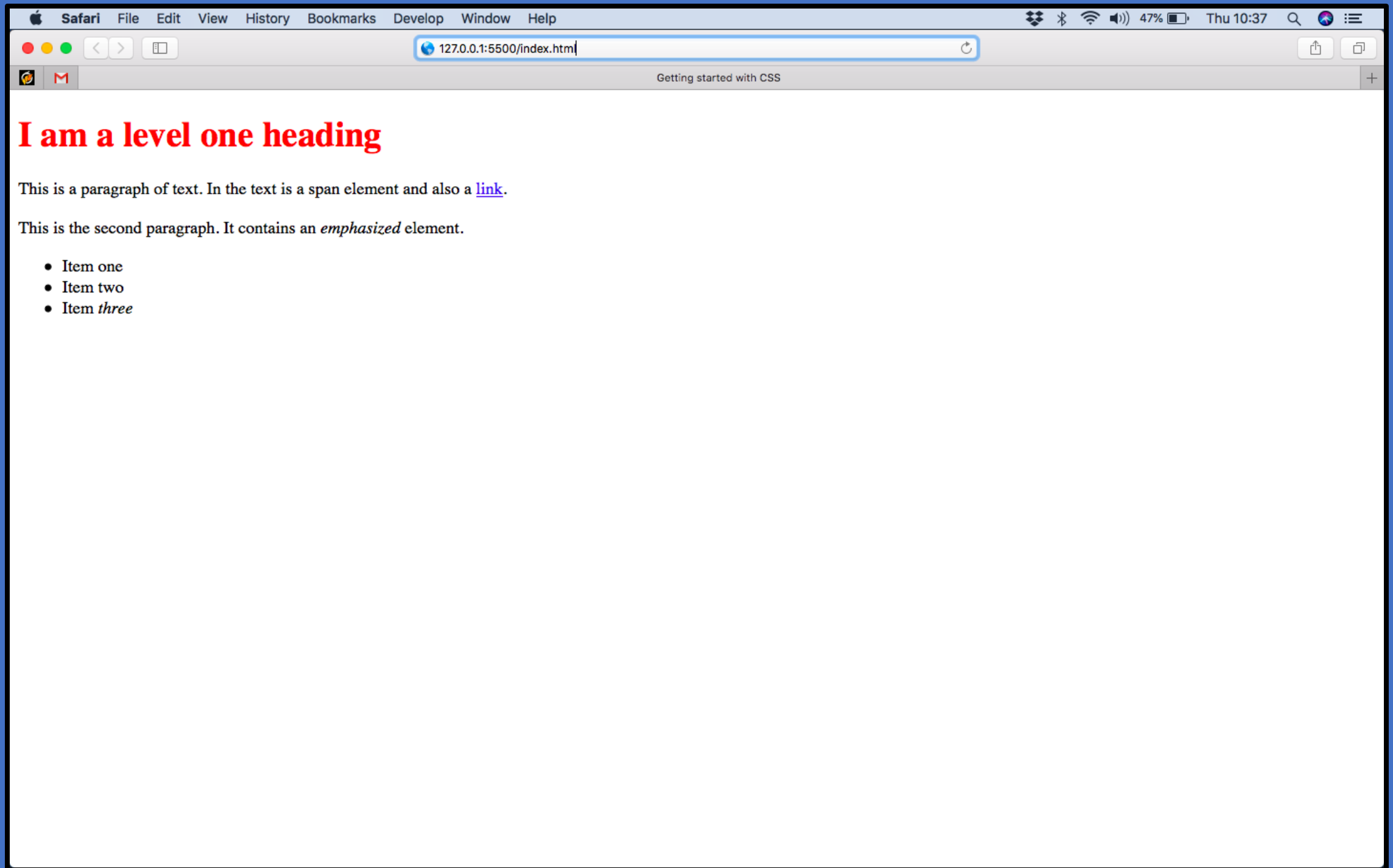
Adding CSS to our document

- Create a file in the same folder as your HTML document and save it as `styles.css`.
- You need to link the web document with the newly created `css` file.

1	<code><link rel="stylesheet" href="styles.css"></code>
---	--

- You can test that the CSS works by adding a rule to `styles.css`.

1	<code>h1 {</code>
2	<code> color: red;</code>
3	<code>}</code>



I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

SafariFileEditViewHistoryBookmarksDevelopWindowHelpcodepen.ioA Pen by Captain Anonymous

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HTML

```
1 <body>
2   <h1>I am a level one heading</h1>
3
4   <p>This is a paragraph of text. In the text is a <span>span element</span>
5   and also a <a href="http://example.com">link</a>.</p>
6
7   <p>This is the second paragraph. It contains an <em>emphasized</em> element.</p>
8
9   <ul>
10    <li>Item one</li>
11    <li>Item two</li>
12    <li>Item <em>three</em></li>
13  </ul>
14
15 </body>
```

CSS

```
1 h1 {
2   color: red;
3 }
```

JS

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

ConsoleAssets⌘ KeysSlack: Where Work Happens

Styling HTML elements

Styling HTML elements

- We do this by **targeting** an **element selector**.
 - ✓ **Selector** that **directly** matches an **HTML element name**.
- To **target all paragraphs** in the **document** you would use the selector `p`.
- To turn **all paragraphs green**:

```
1 p {  
2   color: green;  
3 }
```

Styling HTML elements

- You can **target multiple selectors** at once, by **separating the selectors** with a **comma**.
- If I want **all paragraphs** and **all list items** to be **green**:

1	p, li {
2	color: green;
3	}

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codepen.io

A Pen by Captain Anonymous

Untitled
Captain Anonymous

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HTML

```
1<h1>I am a level one heading</h1>
2
3
4<p>This is a paragraph of text. In the text is a <span>span element</span>
5and also a <a href="http://example.com">link</a>.</p>
6
7<p>This is the second paragraph. It contains an <em>emphasized</em> element.</p>
8
9<ul>
10<li>Item one</li>
11<li>Item two</li>
12<li>Item <em>three</em></li>
13</ul>
14
15</body>
16
```

CSS

```
1h1 {
2  color: red;
3}
4
5p, li {
6  color: green;
7}
```

JS

```
1
```

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

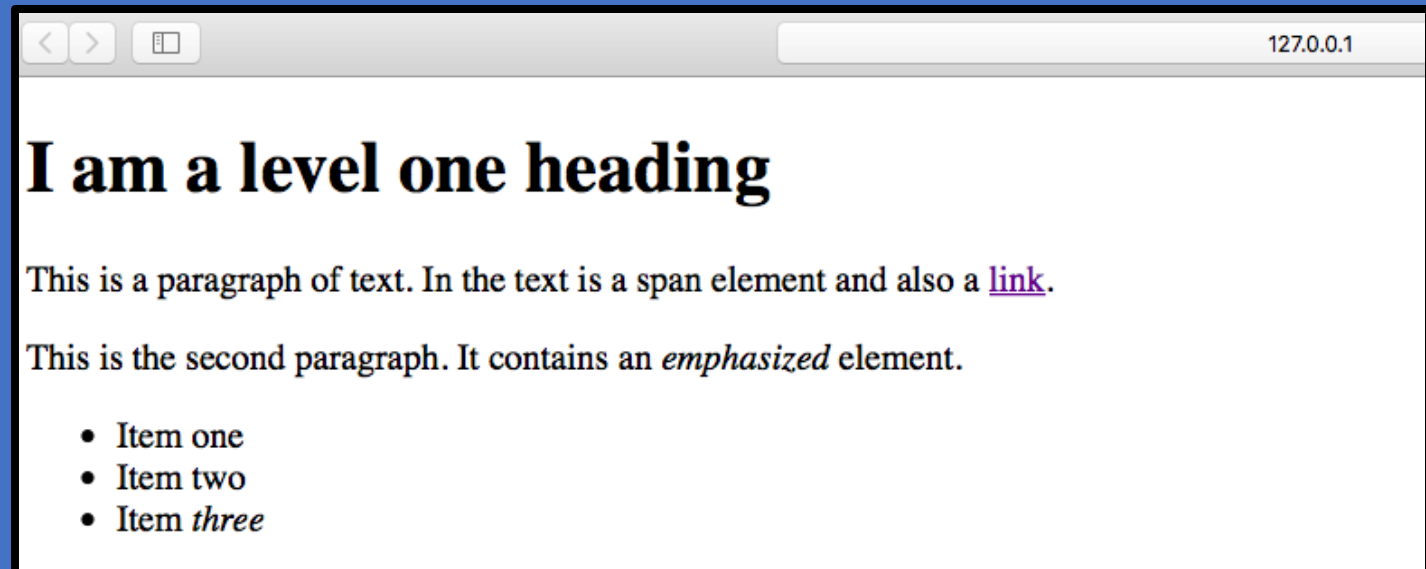
- Item one
- Item two
- Item *three*



Changing the default behavior of elements

Changing the default behavior of elements

- When the **browser** applies **default styling**, headings are large and **bold** and our **list** has **bullets**.
 - ✓ **Browsers** have **internal stylesheets** containing **default styles**, which they **apply** to all pages.
- All **modern browsers** display **HTML content** by default in pretty much the same way.



Changing the default behavior of elements

- However, you will often want something other than the choice the browser has made.
 - ✓ For example, if you decide you don't want those bullets in a list, you can remove them:

```
1 li {  
2   list-style-type: none;  
3 }
```

SafariFileEditViewHistoryBookmarksDevelopWindowHelpcodepen.ioA Pen by Captain Anonymous

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HTML

```
1<h1>I am a level one heading</h1>
2
3
4<p>This is a paragraph of text. In the text is a <span>span element</span>
5and also a <a href="http://example.com">link</a>.</p>
6
7<p>This is the second paragraph. It contains an <em>emphasized</em> element.</p>
8
9<ul>
10<li>Item one</li>
11<li>Item two</li>
12<li>Item <em>three</em></li>
13</ul>
14
15</body>
16
```

CSS

```
1h1 {
2  color: red;
3}
4
5p, li {
6  color: green;
7}
8
9li {
10  list-style-type: none;
11}
```

JS

```
1
```

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

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I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*



```
li{  
  list-style-type: square;  
}
```

list-style-type

Web technology for developers › CSS: Cascading Style Sheets › list-style-type

English ▼

On this Page

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Related Topics

[CSS](#)

[CSS Reference](#)

[CSS Lists and Counters](#)

▼ [Guides](#)

[Consistent list indentation](#)

[Using CSS counters](#)

The **list-style-type** CSS property sets the marker (such as a disc, character, or custom counter style) of a list item element.



CSS Demo: list-style-type

Reset

```
list-style-type: space-counter;
```

```
list-style-type: disc;
```

```
list-style-type: circle;
```

```
list-style-type: "\1F44D"; // thumbs up sign
```

NASA Notable Missions

- Apollo
- Hubble
- Chandra
- Cassini-Huygens

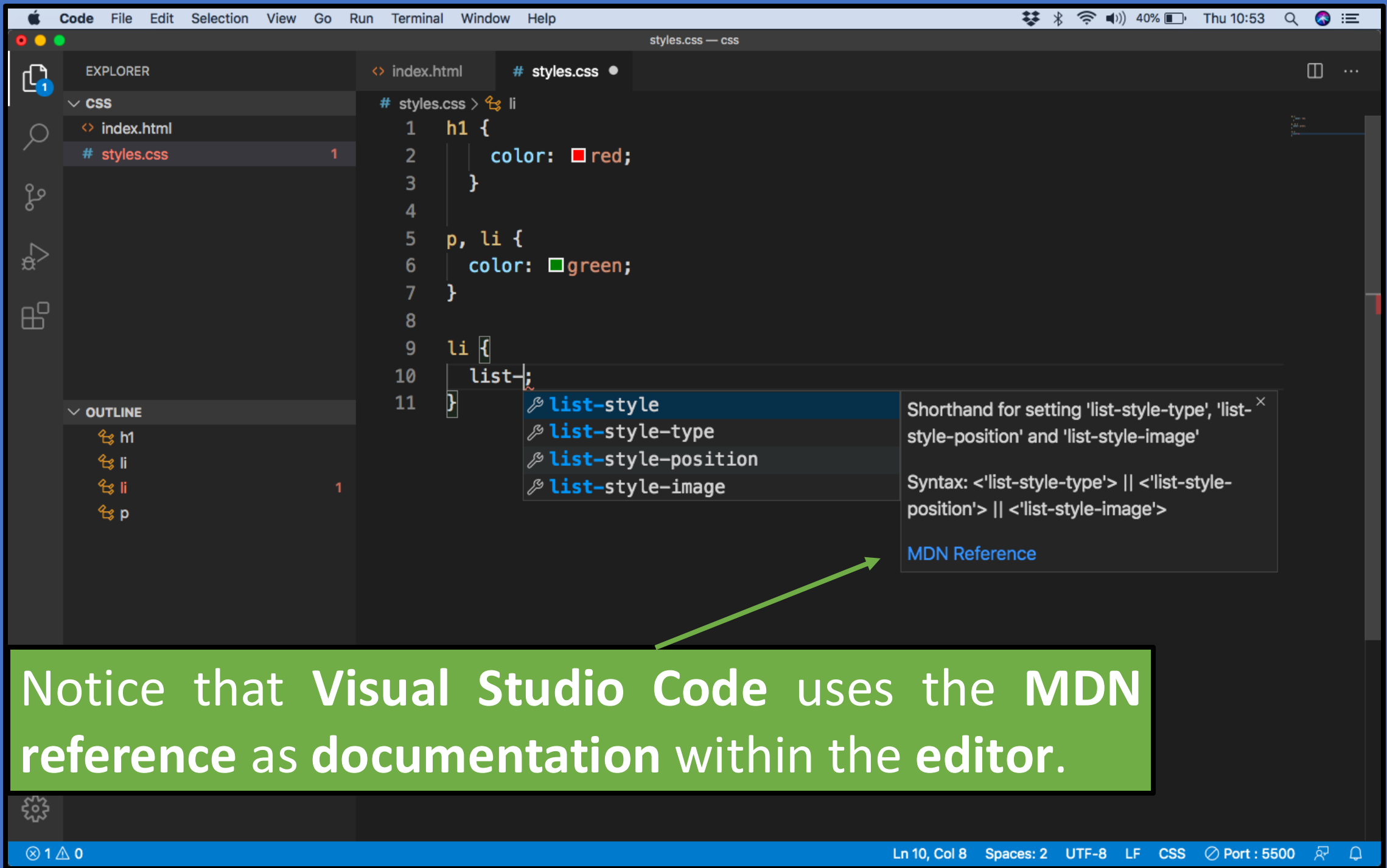
space-counter is defined with [@counter-style](#)

The **color** of the marker will be the same as the computed color

Only a few elements (`<li>` and `<summary>`) have a default value of `display: list-item`

<https://mzl.la/2WMUp53>

Integration of Visual Studio Code with MDN reference documentation



EXPLORER

CSS

index.html

styles.css 1

OUTLINE

html

body

h1

p

a

span

p

em

ul

li

li

li

em

head

link

meta

title

index.html

styles.css

index.html > html > body

6 <title>Getting started with CSS</title>

7 </head>

8

9 <body>

10

11 <h1>I am a level one heading</h1>

12

13 <p>This is a paragraph of text. In the text is a span element

14 and also

15

16 <p>Th

17 eleme

18

19

20 Item two

21 Item three

22

23

24 </body>

25

26 </html>

The li element represents a list item. If its parent element is an ol, ul, or menu element, then the element is an item of the parent element's list, as defined for those elements. Otherwise, the list item has no defined list-related relationship to any other li element.

MDN Reference

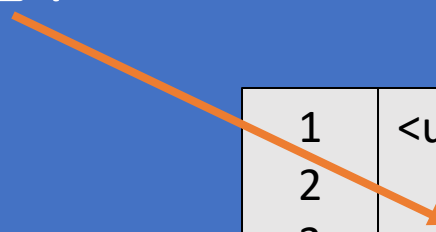
Adding a class

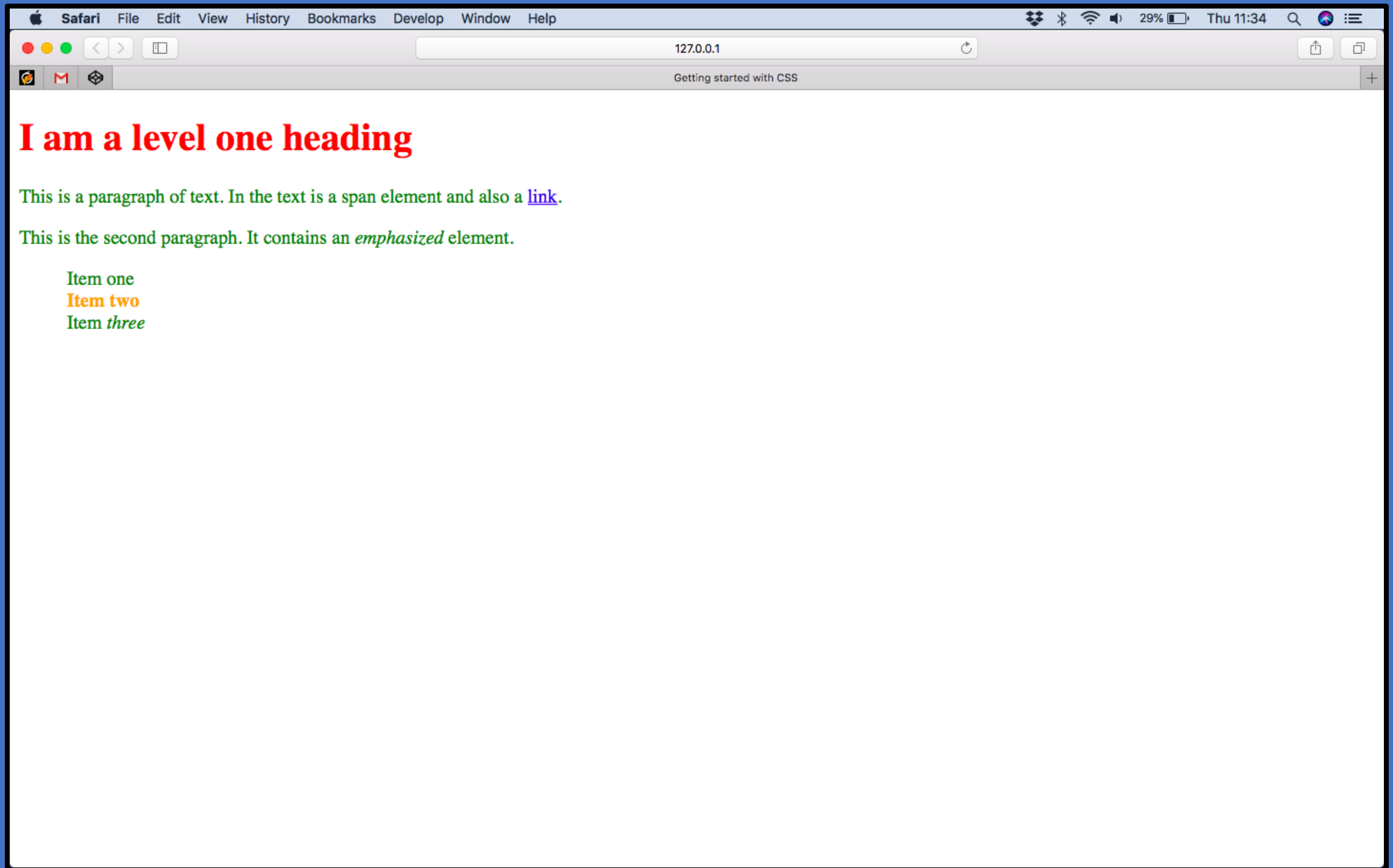
Adding a class

- You will need to find a way to select a subset of the elements without changing the others.
- The most common way to do this is to add a class to your HTML element and target that class.
 - ✓ We named the class “special”.

1	.special {
2	color: orange;
3	font-weight: bold;
4	}

1	
2	Item one
3	<li class="special">Item two
4	Item three
5	





I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

Item one

Item two

Item *three*

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Captain Anonymous

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HTML11 unsaved changes X

```
1<!-- ... -->
2<h1>I am a level one heading</h1>
3
4<p>This is a paragraph of text. In the text is a <span>span
  element</span>
5and also a <a href="http://example.com">link</a>.</p>
6
7<p>This is the second paragraph. It contains an <em>emphasized</em>
  element.</p>
8
9<ul>
10<li>Item one</li>
11<li class="special">Item two</li>
12<li>Item <em>three</em></li>
13</ul>
14
```

CSS

```
1
2
3
4
5p, li {
6  color: green;
7 }
8
9li {
10  list-style-type: none;
11 }
12
13.special {
14  color: orange;
15  font-weight: bold;
16 }
```

JS

```
1
```

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

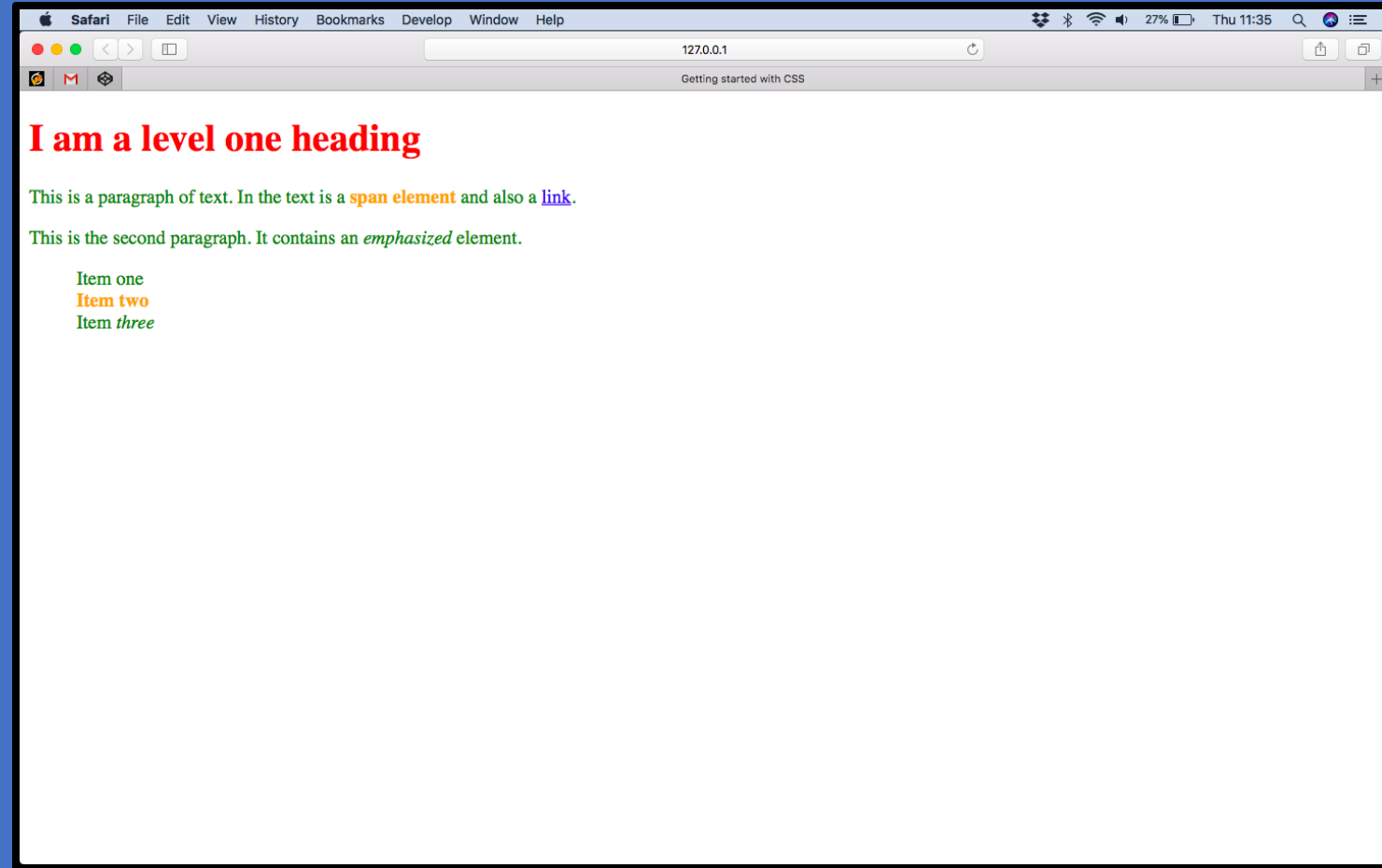
This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two
- Item *three*

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Adding a class

- You can apply the previous class named "special" to any element on your page that you want to have the same look as this list item.
- For example, you might want the `<span>` in the paragraph to also be orange and bold.



Adding a class

- Sometimes you will see rules with a selector that lists the **HTML element selector along with the class**
 - ✓ Meaning "target any `li` and `span` elements that has a class named `special`".

1	<code>li.special,</code>
2	<code>span.special {</code>
3	<code>color: orange;</code>
4	<code>font-weight: bold;</code>
5	<code>}</code>

**Styling things based on
their location in a document**

Styling things based on their location

- There are a **number of selectors** that can **help**, but for **now** we will look at just a **couple**.
- To **select** only an `<em>` that is **nested** inside an `<li>` **element** I can use a **selector called the descendant combinator**.
 - ✓ A **descendant combinator** takes the form of a **space between two other selectors**.

1	li em {
2	color: rebeccapurple;
3	}

I am a level one heading

This is a paragraph of text. In the text is a **span element** and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two**
- Item *three*

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Captain Anonymous

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HTML

```
1<h1>I am a level one heading</h1>
2
3
4<p>This is a paragraph of text. In the text is a <span>span
  element</span>
5and also a <a href="http://example.com">link</a>.</p>
6
7<p>This is the second paragraph. It contains an <em>emphasized</em>
  element.</p>
8
9<ul>
10<li>Item one</li>
11<li class="special">Item two</li>
12<li>Item <em>three</em></li>
13</ul>
14
```

CSS 12 unsaved changes

```
1
2
3
4
5
6
7
8
9li {
10  list-style-type: none;
11 }
12
13.special {
14  color: orange;
15  font-weight: bold;
16 }
17
18li em {
19  color: rebeccapurple;
20 }
```

JS

I am a level one heading

This is a paragraph of text. In the text is a span element and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

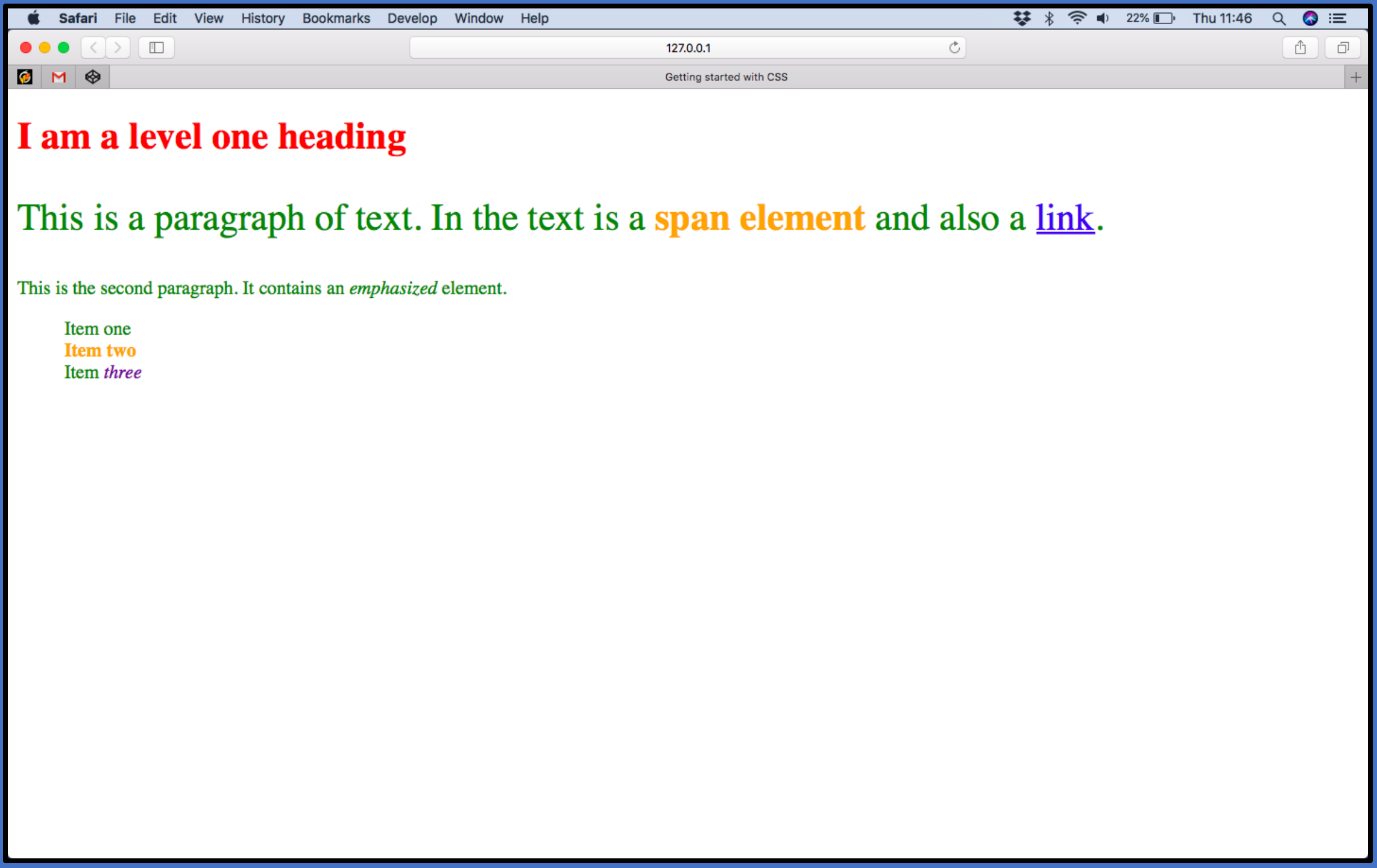
Item one
Item two
Item *three*



Styling things based on their location

- The following example styles a paragraph when it comes directly after a heading at the **SAME** hierarchy level in the **HTML**.
- To do so place a **+** (an adjacent sibling combinator) between the selectors.

1	<code>h1 + p {</code>
2	<code>font-size: 200%;</code>
3	<code>}</code>



I am a level one heading

This is a paragraph of text. In the text is a **span element** and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- Item two**
- Item *three*

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HTML

```
1<h1>I am a level one heading</h1>
2
3
4<p>This is a paragraph of text. In the text is a <span>span
  element</span>
5and also a <a href="http://example.com">link</a>.</p>
6
7<p>This is the second paragraph. It contains an <em>emphasized</em>
  element.</p>
8
9<ul>
10<li>Item one</li>
11<li class="special">Item two</li>
12<li>Item <em>three</em></li>
```

CSS13 unsaved changes X

```
14color: orange;
15font-weight: bold;
16}
17
18li em {
19color: rebeccapurple;
20}
21
22h1 + p {
23font-size: 200%;
24}
```

JS

```
1
```

I am a level one heading

This is a paragraph of text. In the text is a span element and also a link.

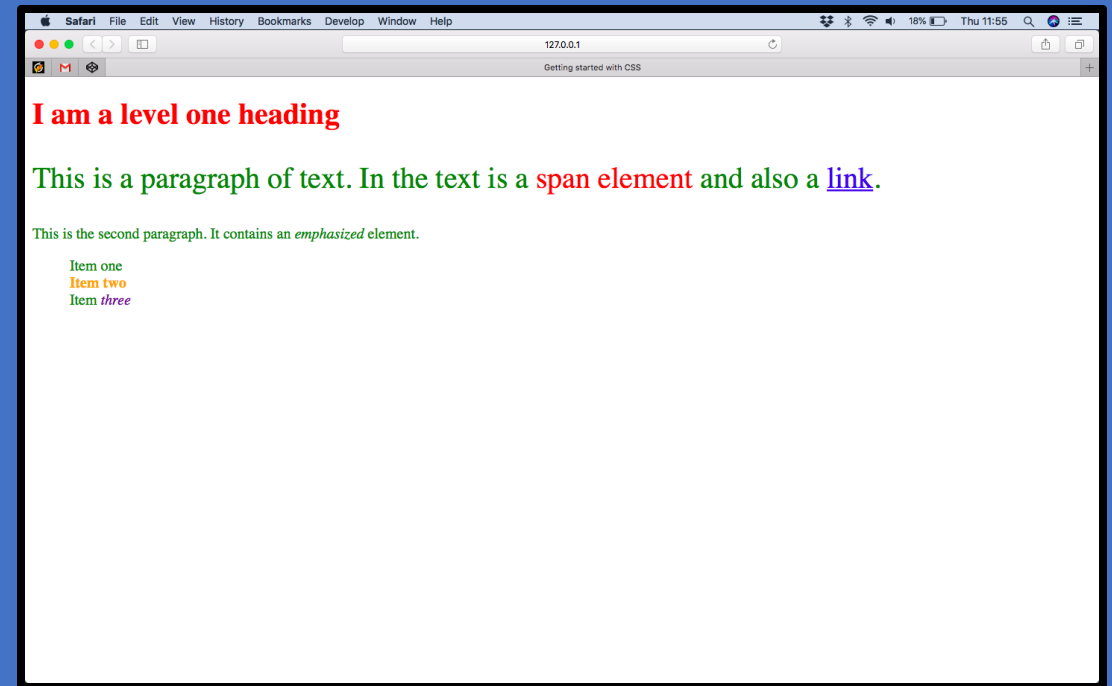
This is the second paragraph. It contains an emphasized element.

Item one
Item two
Item three

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Styling things based on their location

- Try adding a rule to make a `span` red, if it is inside a paragraph.
- You will know if you have it right as the `span` in the first paragraph will be red, but the one in the first list item will not change color.

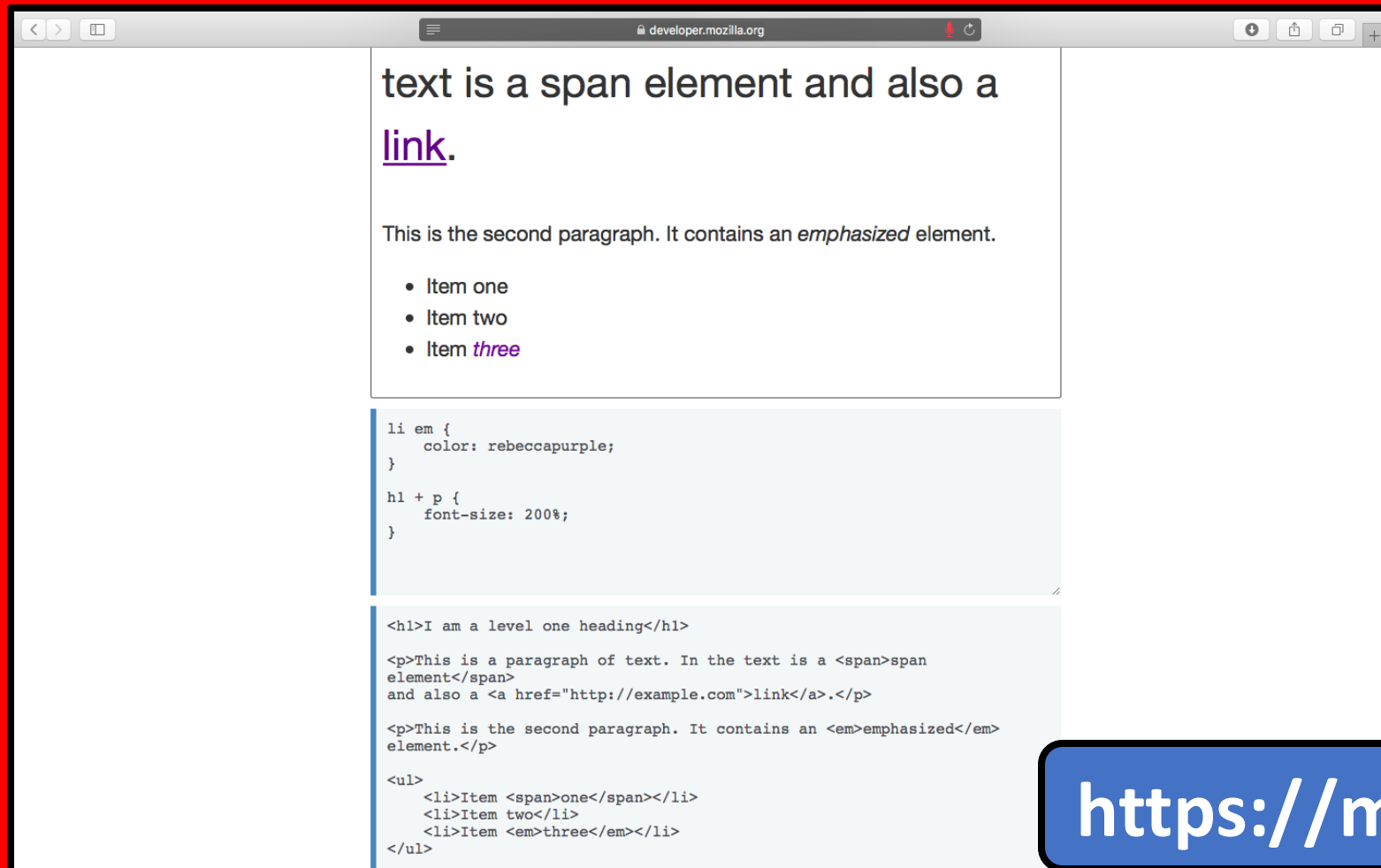


Note

Note: As you can see, **CSS** gives us **several ways** to **target elements**, and we've **only scratched the surface** so far! We will be **taking a proper look** at all of these **selectors** and **many more** in our **Selectors lecture** later on in the **course**.

**Active learning:
Location based styling**

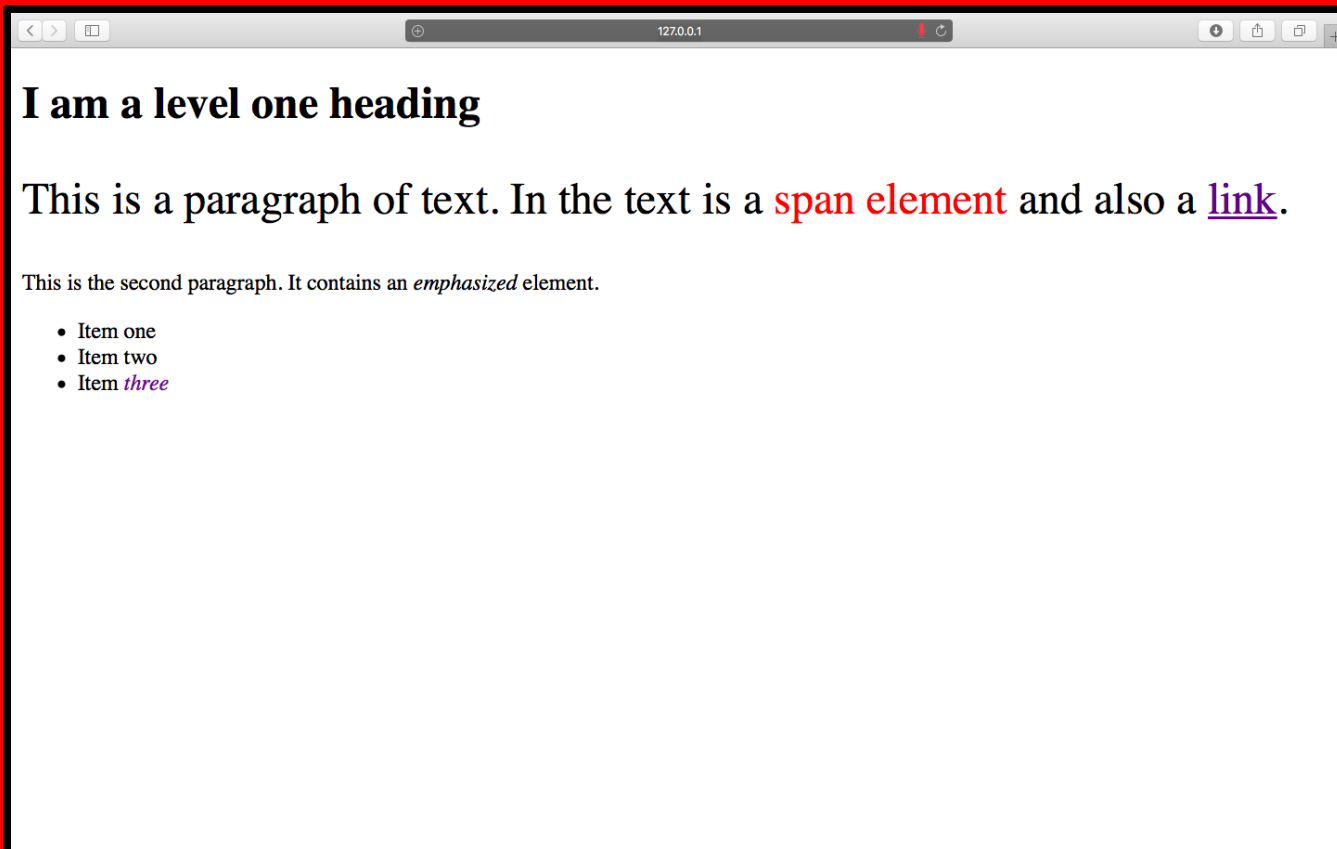
Try adding a rule to make a `span` red, if it is inside a paragraph.



<https://mzl.la/2YvKaCv>

Solution

The `span` in the first paragraph will be **red**, but the one in the first list item will not change color.



Styling things based on state

Styling things based on state

- A straightforward example is when styling links.
- The `<a>` (anchor) element has different states depending on whether it is **unvisited**, **visited**, being **hovered over**, **focused** via the **keyboard**, or in the process of being **clicked** (activated).
- You can use **CSS** to **target** these **different states**.

Styling things based on state

- This CSS example styles unvisited links **pink** and visited links **green**.

1	<code>a:link {</code>
2	<code> color: pink;</code>
3	<code>}</code>
4	
5	<code>a:visited {</code>
6	<code> color: green;</code>
7	<code>}</code>

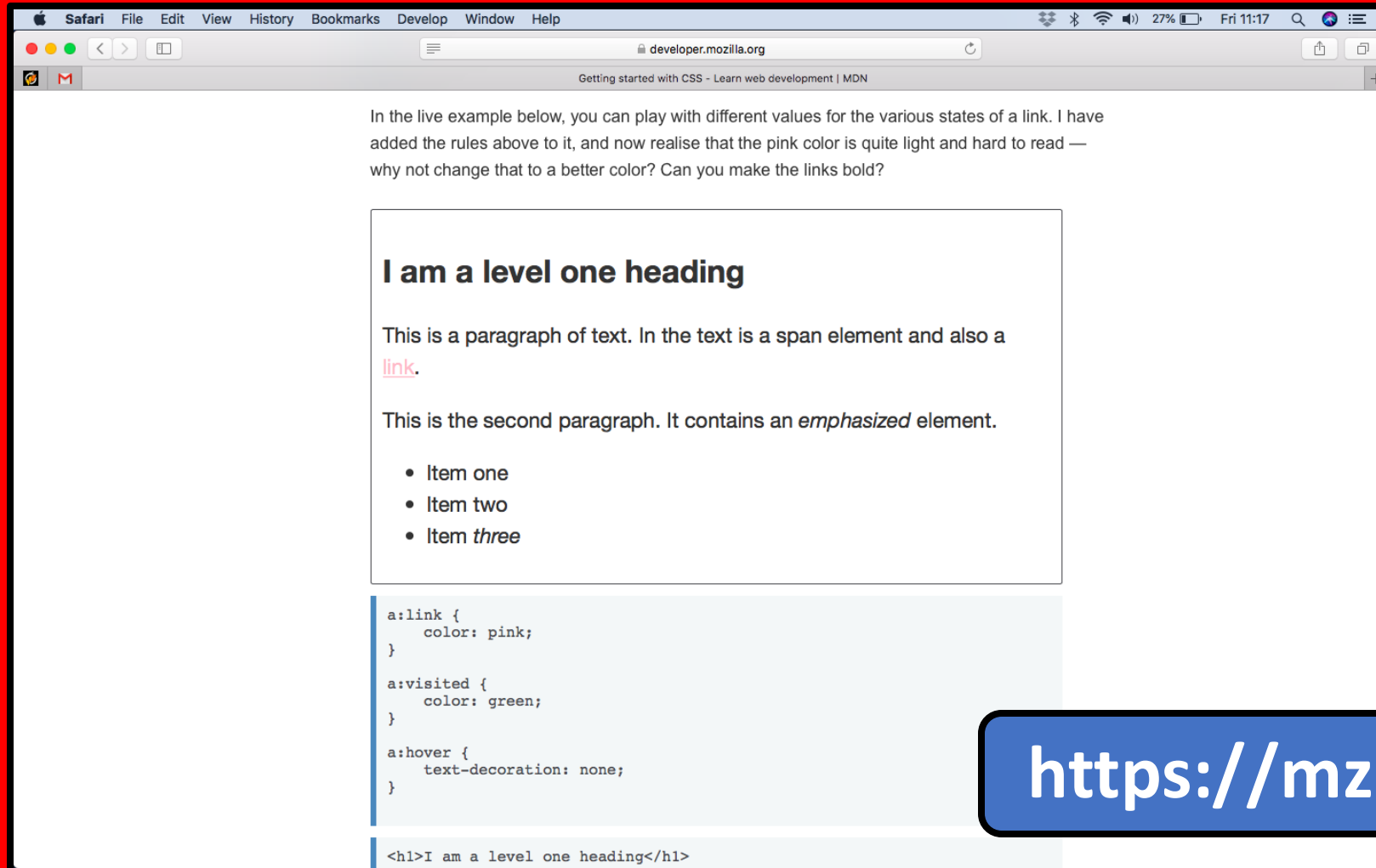
Styling things based on state

- When the user **hovers over it**, we can, for **example**, **remove the underline**.
 - ✓ **Avoid** doing this for **accessibility reasons**.

```
1 a:hover {  
2   text-decoration: none;  
3 }
```

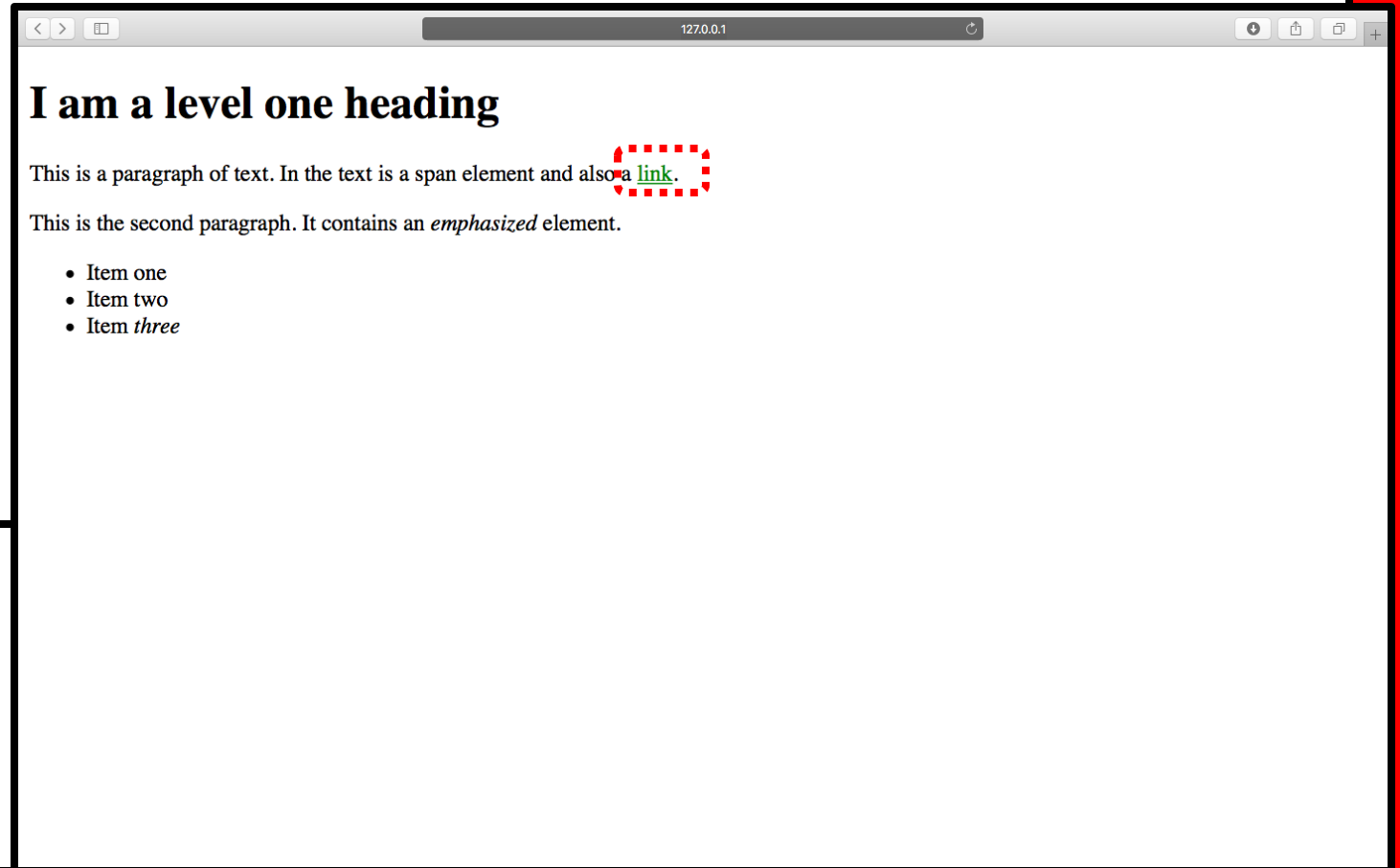
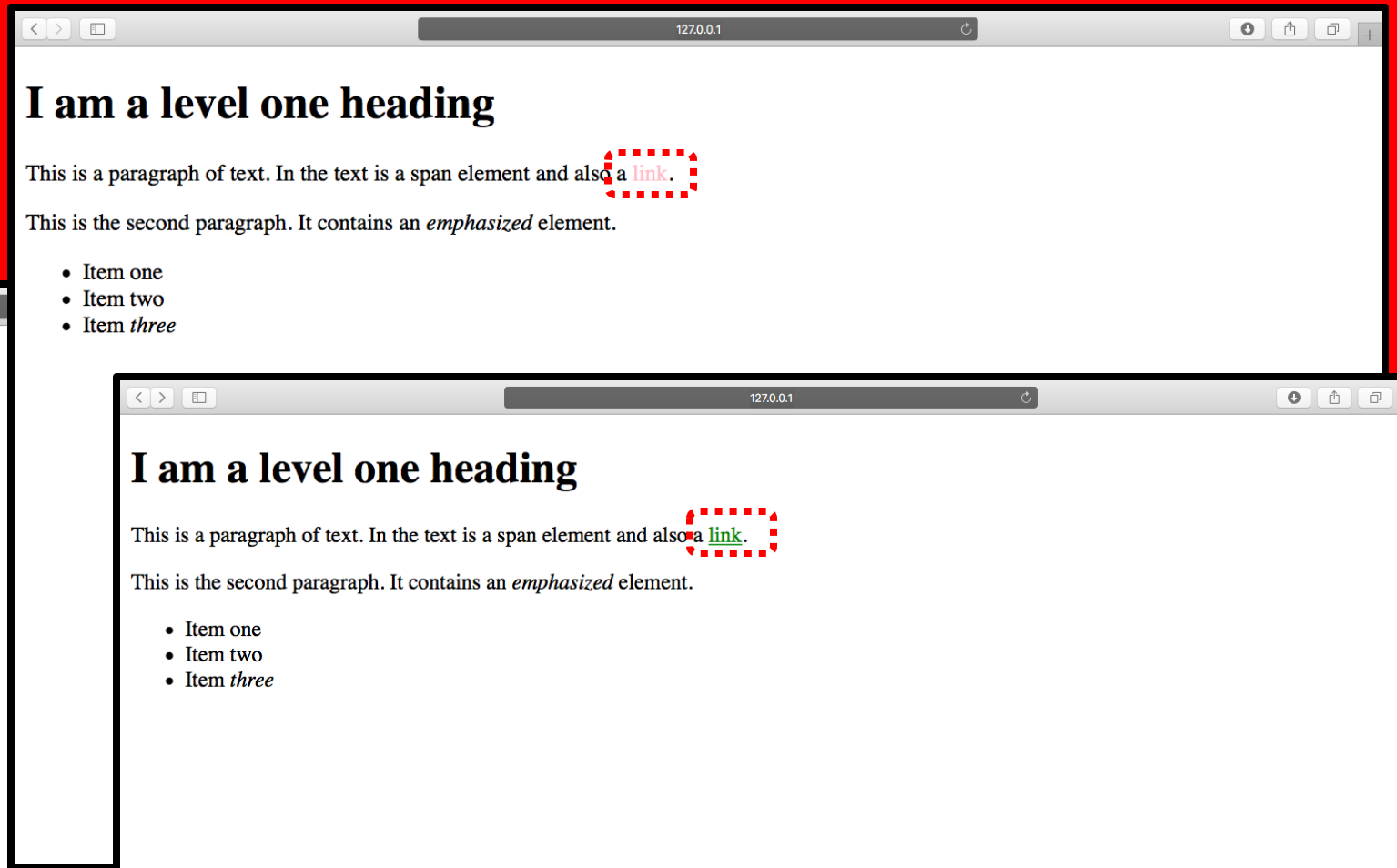
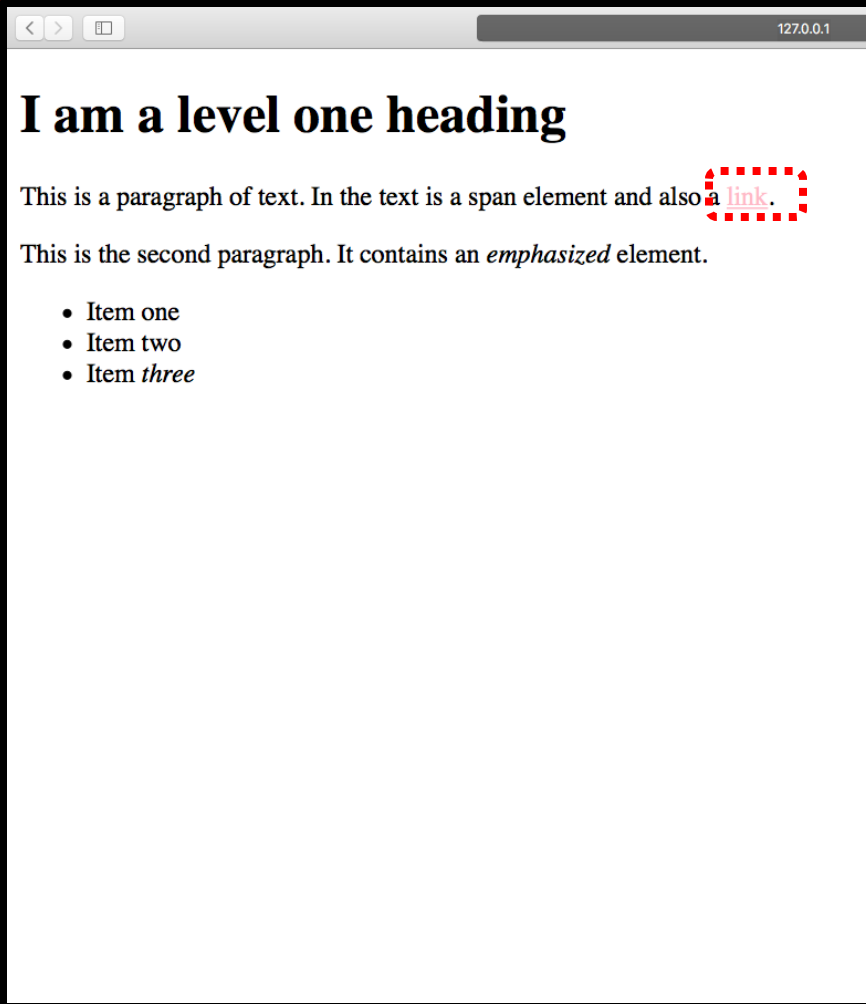
**Active learning:
State based styling**

Go to the link indicated below for the Active Learning statement



<https://mzl.la/2EPjYeT>

Solution



Note: Your **visitors** may well be on a **computer** with a **mouse** or **trackpad**, or a **phone** with a **touchscreen** or they might be using a **screenreader**, which reads out the content of the document, or they may need to use much **larger text**, or be navigating the **site** using the **keyboard** only. A **plain HTML document** is generally **accessible to everyone** — as you start to **style** that **document** it is **important** that you **don't make it less accessible**.

Combining selectors and combinators

Combining selectors and combinators

- You can **combine multiple selectors and combinators together**.

```
1  /* selects any <span> that is inside a <p>, which is inside an <article> */  
2  article p span { ... }  
3  
4  /* selects any <p> that comes directly after a <ul>, which comes directly after an <h1> */  
5  h1 + ul + p { ... }
```

Combining selectors and combinators

- You can combine multiple types together, too.
- This will **style** any **element** with a **class named** `special` inside a `<p>`, just after an `<h1>`, which is inside a `<body>`.

```
1 body h1 + p .special {  
2   color: yellow;  
3   background-color: black;  
4   padding: 5px;  
5 }
```

I am a level one heading

This is a paragraph of text. In the text is a **span element** and also a [link](#).

This is the second paragraph. It contains an *emphasized* element.

- Item one
- **Item two**
- Item *three*

```
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3   background-color: black;  
4   padding: 5px;  
5 }
```

Questions?